CLASS 8 — SCIENCE · FORMULA SHEET

Chapter 12: **How Nature Works in Harmony** | Quick revision, one page

Components of an ecosystem

- Biotic: producers, consumers, decomposers. Abiotic: sunlight, air, water, soil, temperature.
- Producers (green plants) make food by photosynthesis; the Sun is the ultimate source of energy.

Consumers

- Herbivores eat plants; carnivores eat animals; omnivores eat both.
- Primary consumers are herbivores; secondary consumers eat primary consumers; the top carnivore ends the chain.

Chains and webs

- Food chain example: grass → grasshopper → frog → snake → eagle (arrow points towards the eater).
- Food web: many interlinked food chains; more stable than a single chain.

Energy and balance

- Energy flows one way: producers → consumers; part is lost as heat at each step.
- Number of organisms generally decreases from producers to top carnivores.
- Decomposers recycle nutrients; removing one species can upset the balance.

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